

DEVOPS

1. What is DevOps?

DevOps = Development + Operations

It is a way of working where developers and IT operations teams work together to **build, test, deploy, and maintain software faster and better.**

Main goals of DevOps:

- Faster software delivery
- Better collaboration between teams
- Automated testing and deployment
- Fewer errors
- Continuous improvement

Example tools: Git, Jenkins, Docker, Kubernetes

2. Advantages of DevOps

- **Faster delivery** – software updates are released quickly
 - **Better teamwork** – developers and operations work together
 - **Automation** – reduces manual work
 - **Higher quality** – bugs are found early
 - **Reliability** – systems are more stable
 - **Quick problem fixing** – faster recovery from failures
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3. What is Agile?

Agile is a project management and software development method where work is done in **small parts (iterations)** instead of all at once.

Teams deliver small improvements regularly and adapt based on feedback.

Agile principles:

- Customer collaboration
 - Fast delivery
 - Flexibility to changes
 - Team communication
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4. What is Scrum?

Scrum is one of the most popular **Agile frameworks**.

Work is divided into short time periods called **Sprints** (usually 2–4 weeks).

In each sprint, the team plans, builds, tests, and delivers a small working product.

Scrum activities:

- Sprint Planning
 - Daily Stand-up Meeting
 - Sprint Review
 - Sprint Retrospective
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5. Who is Product Owner (PO)?

Product Owner (PO) represents the customer and decides **what features should be built**.

Responsibilities:

- Defines product requirements
- Prioritizes tasks (Product Backlog)
- Talks with stakeholders/customers
- Ensures the team builds the right product

Simple: *PO decides **what** to build.*

6. Who is Scrum Master?

Scrum Master helps the team follow Scrum properly.

Responsibilities:

- Removes obstacles for the team
- Conducts Scrum meetings
- Helps team improve workflow
- Protects team from distractions

Simple: *Scrum Master helps **how** the team works.*

7. Advantages of Agile

- **Flexible to changes**
- **Faster delivery of features**
- **Better customer satisfaction**
- **Improved teamwork**
- **Early bug detection**
- **Continuous feedback and improvement**